Faster, Higher, Further Apart

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Abstract

In light of the recent decline in success of the German Athletics Federation (DLV) at the international level, this report examines the structural development of the extended national elite. To this end, we compiled the German athletics rankings from 2001 to 2025 into a dataset of over 238,000 entries. Using this dataset, we are investigating the impact of the COVID-19 pandemic and long-term shifts in performance spreads. We utilize Scores from World Athletics organization to make performances comparable across disciplines. Our results show a significant decline in performance during the pandemic (-2.14% median) and an increasing polarization of the extended elite. While the absolute top performers improved, the extended elite declined. This decrease in performance density is particularly evident among men and in technical disciplines and suggests structural deficiencies in coaching presence and internal competition. All code and data are available here, and the results can be explored interactively via our Streamlit dashboard.

1. Introduction

As the world's largest athletics association, the Deutscher Leichtathletik-Verband (DLV) is tasked with facilitating peak individual performance and maintaining an internationally competitive national team (Deutscher Leichtathletik-Verband, n.d.a). However, zero medals at the 2023 World Athletics Championships in Budapest and a historic low medal count at the 2024 Paris Olympics have left the organization in a sobering position. Although individual athletes still occasionally reach the podium, the declining success with regard to Olympic performances (Fremerey et al., 2024) raises questions about the fundamental strength of the national elite and extended elite.

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While medals are the most visible metric of success, they reflect only the peak of elite performance and offer little insight into how performance is distributed within the elite level. To examine potential structural shifts, we analyze the top 35 rankings from 2001 to 2025 based on two research questions. First, we exploratively assess the impact of the COVID-19 pandemic as a short-term disruption to the performance development. Second, motivated by observed performance drop-offs beyond the top positions, we assess whether the performance gap within the elite has widened over time. Such a trend could indicate reduced internal competition, which can weaken international success, as elite development relies on strong competitive environments at the club level (Tihanyi, 2001, cited in Green & Houlihan, 2005).

Our analytical approach is detailed in Section 2, where we describe the data processing and justify the analytical framework. Building on this, we will analyze performance metrics to identify structural trends and evaluate our research questions in Section 3. Finally, Section 4 discusses the implications of these findings.

2. Data and Methods

2.1. Data Acquisition and Standardization

The data were retrieved from the official archive of the DLV, documenting top performances by German athletes in national and international competitions across major athletic disciplines (running, hurdles and steeplechase, jumping, throwing, and combined events) from 2001 to 2025. Performances are classified by sex (male, female) and age group (U14 to adults).

Data were extracted from various PDF files using regular expressions to handle formatting variations across years. Outliers and entry inconsistencies (e.g., irregular time formats) were corrected via external sources where possible. Minor data gaps from parsing errors were manually completed by referencing the original files. Non-recurrent distances (e.g., 7.5 km youth events) were excluded, resulting in a final dataset of 238,187 entries.

We standardized discipline names, birth years and converted performance marks into numeric values (seconds for track events and meters for field events) to allow statistical analyses. To enable comparisons across disciplines, all performance marks were transformed into World Athletics scoring points (IAAF scores). These scores map heterogeneous results onto a common scale theoretically ranging from 0 to 1400, where world records generally exceed 1300 points (World Athletics, 2025). As no official formula for score computation is publicly available, we implemented reverse-engineered, discipline-specific quadratic functions based on 2025 scoring tables (Chen, 2019). The following analyses are based on these standardized IAAF scores.

2.2. Data Selection and Grouping

To ensure a representative sample of elite performance, the analyses include the top 35 national results per year, sex, and age group. This threshold establishes a truncated distribution, focusing specifically on the high-performance tail of athletic performances. A "Group" is defined as the combination of sex, age category, and discipline, consistent with official leaderboards. Although the dataset covers a broader range of age categories, the analyses focus on the U18, U20, and Adult groups, as these are the primary age groups competing at international championship level. To facilitate cross-disciplinary comparison, disciplines were aggregated into four primary clusters based on current team selection criteria (Deutscher Leichtathletik-Verband, 2025): sprints, runs, throws, and jumps. Due to regulatory changes in hurdle heights and throwing weights, affected disciplines were excluded from median and rank-specific analyses but retained for gap evaluations to maintain longitudinal comparability.

2.3. Analysis Plan: Impact of COVID-19

To investigate the impact of the COVID-19 pandemic and to get an overall impression of the data, an exploratory analysis was conducted. A heatmap displaying the number of competitive events per month and year was used to examine seasonal competition patterns and disruptions in event frequency and data density during the pandemic period. Events were defined as performances recorded on the same date and in the same city.

To isolate the effect of the pandemic year, performances were grouped into two cohorts: a pre-pandemic baseline comprising pooled IAAF scores from 2015–2019, and a pandemic cohort consisting of all IAAF scores recorded in 2020. General performance shifts were assessed using the relative percentage difference between period medians. Changes in performance density were evaluated using the interquartile range (IQR). The median and IQR were selected as robust measures of central tendency and dispersion, as our data are non-normally distributed and may include extreme values. The primary indicator of distributional change was the relative percentage change in median and IQR between 2020 and the baseline period.

Applicaple to the whole study, a positive percentage change in the median represents performance improvement, while a positive change in IQR indicates a widening performance gap. We determined the statistical significance of median shifts using the Mann-Whitney U test, as it does not require normal distribution. To evaluate changes in performance density, we utilized bootstrap resampling (B=10,000), which is particularly robust for our specific sample sizes for the following performance gap measurement with N=15. Significant changes in density were identified when the 95% confidence intervals (CIs) for the variance differences excluded zero.

2.4. Analysis Plan: Widening Performance Gap

To analyze our second research question, we hypothesize that the difference between the best and worst performances in our data set has increased over the past two decades. By comparing two five-year periods (2001–2005 vs. 2021–2025), annual variability will be reduced. The analysis is conducted at two levels. First, overall trends in performance level and density are examined across all groups to identify general patterns. Second, rank-specific developments are analyzed to assess how performance changes are distributed exactly.

Performance density is quantified using a performance gap measure (G_p), which is defined as the difference between the mean of the top three and the mean of the bottom three performances (ranks 33–35) within each group and period. Averaging across the top three reflects international success standards (podium level), while the bottom three provide a stable lower benchmark. The performance gap is calculated as:

$$G_p = \left(\frac{1}{3|P|} \sum_{y \in P} \sum_{r=1}^{3} x_{y,r} \right) - \left(\frac{1}{3|P|} \sum_{y \in P} \sum_{r=33}^{35} x_{y,r} \right), \tag{1}$$

where $x_{y,r}$ denotes the performance of rank r in year y, and $|P| = 5$ represents the number of years per period. Changes in the performance gap are reported as the relative percentage difference between the two periods. Statistical significance was determined, as detailed in Section 2.3, using the Mann-Whitney U test for the median and bootstrap resampling for the performance gap.

To examine rank-specific developments, arithmetic means were calculated for each rank across all groups for both periods. Sample sizes per rank, even when differentiating by age group and sex, were sufficiently large (minimum $n \geq 80$) to assume a normal distribution within the rank and thus calculate the arithmetic mean. The development of each rank was determined by calculating the percentage change between the start and end periods, normalized to the start period. A positive value indicates an improvement in the average ranking, a negative value a decline.

Due to regulatory changes in hurdle heights and throwing implement weights, particularly affecting U18 and U20 categories (World Athletics, 2020), the respective disciplines were excluded from median- and rank-based analyses. Performance gap analyses were retained, as this metric can be calculated independently of the absolute values.

3. Results

3.1. Impact of COVID-19

The analysis of the distribution of seasonal events in Figure 1 reveals a noticable shift in the timing of the competition calendar. In 2020, there were more events in the second half of the year, indicating a clear departure from the traditional peak in midsummer. This shift towards late summer and autumn is still partially evident in 2021, although the calendar shows a gradual return to the pre-pandemic seasonal norm.

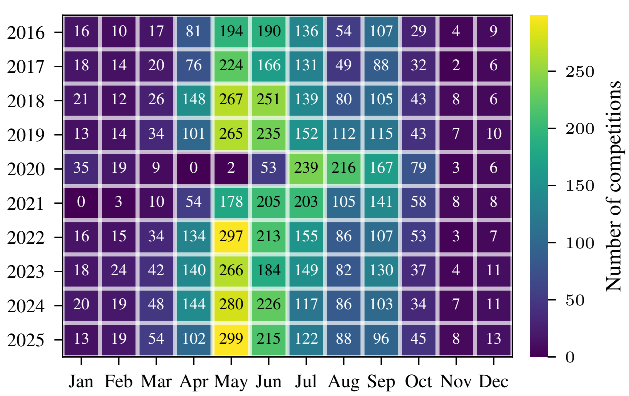

Figure 1. Monthly distribution of competition frequency from 2016 to 2025. Cell values and color intensity represent the absolute number of events recorded per month.

The Mann-Whitney U test confirms a significant decline in performance levels across all athletic categories during the 2020 season compared to the 2015–2019 baseline ($p < 0.05$). This downward shift, captured in Figure 2, resulted in an overall median IAAF score decrease of -2.14% (935 to 915 points). Throws experienced the most pronounced decline (-2.84%), followed by Jumps (-2.45%) and Sprints (-1.94%), while Runs demonstrated the greatest resilience (-0.85%).

Bootstrap analysis revealed significant increases in performance spread, measured by IQR, for Sprint ($+12.04\%$, 95% CI $[4.55, 20.95]$) and for the aggregate dataset (Total: $+5.33\%$, 95% CI $[1.32, 9.40]$). In contrast, the discipline groups Run, Throw, and Jump remained statistically stable, showing no significant change in IQR.

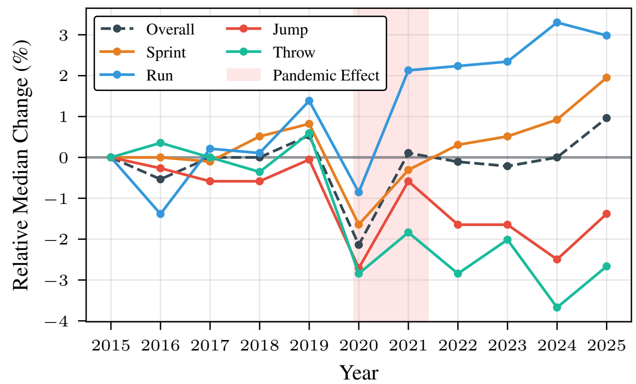

Figure 2. Development of relative median performance change (%) from 2015 to 2025. The year 2015 serves as the baseline (0%). The shaded area highlights the pandemic effect period (2020–2021)

3.2. Widening Performance Gap

Regarding our second research question, Figure 3 illustrates a long-term performance shift across 106 groups between 2001-2005 and 2021-2025. While median trends were nearly balanced (38 improvements vs. 37 declines), performance density decreased significantly: the performance gap widened in 49.06% of groups ($n = 52$) and narrowed in only 6.60% ($n = 7$). Running and sprinting showed the strongest performance, with median improvements occurring two to three times more frequently than declines. In contrast, technical disciplines saw a major decline in median: for throwing, performance decreases were six times more frequent than improvements, while jumping showed a negative ratio of 2.6 to 1.

Looking at the rankings in Figure 4, we see a linear divergence in performance evolution. While the top-ranked athletes improved by an average of 1.61%, the 35th ranked athlete declined by 1.51%. The transition to negative growth occurred at rank 22. Adult women were the notable exception to this trend, showing broad progress, with ranks 3 through 35 improving more than the top two positions. In contrast, the general decline was driven by male youth categories. U18 and U20 men exhibited the earliest drop-offs, with regression starting at ranks 6 and 9, respectively.

4. Discussion & Conclusion

Our analysis of the years during the pandemic reveals a difference between disciplines. Presumably, the pandemic created uncertainty regarding training and future competitions, which likely disrupted targeted preparation. Training

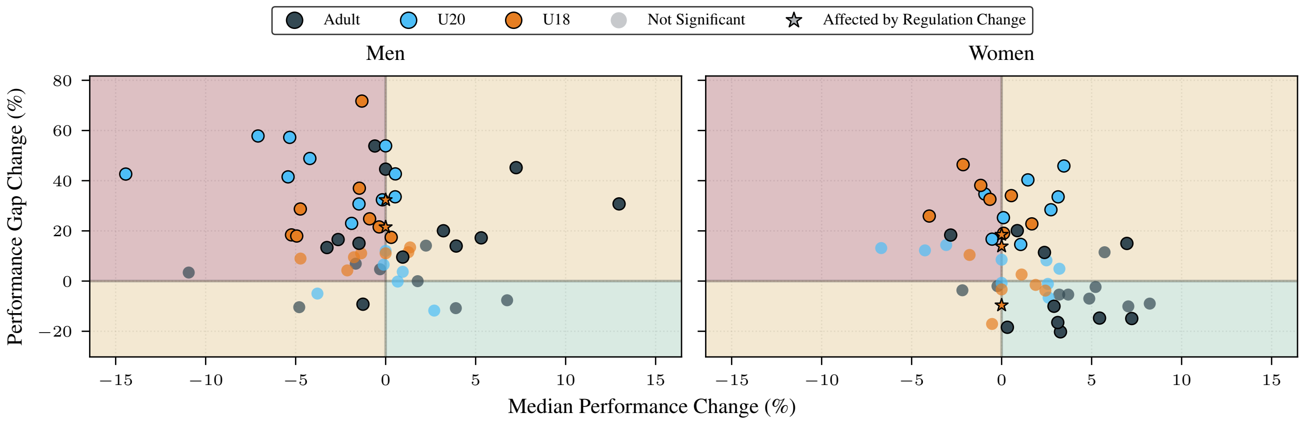

Figure 3. Change in median performance vs. performance gap (2001–2005 to 2021–2025, in %) for men and women across age groups. Marker transparency denotes statistical significance for performance gap (p ¡ 0.05). Median changes for disciplines with regulatory updates are fixed at zero due to restricted comparability. A positive change in the performance gap indicates an increased distance between the top three and the lower-ranked performances (ranks 33–35).

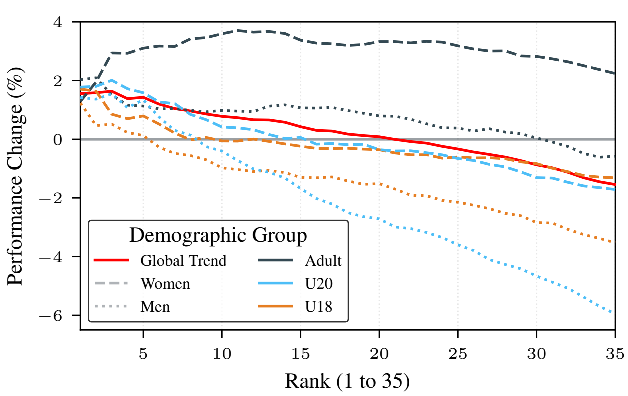

Figure 4. Average rank-specific performance development (2001–2005 vs. 2021–2025, in %), by age group and sex.

facility closures led to a measurable drop in performance in the technical disciplines. Yet, running disciplines may have improved thanks to carbon technology (Willwacher et al., 2023) and uninterrupted training blocks (Venkatesan, 2022). Women, in contrast to men, show a catching-up effect, suggesting that professionalization is not yet complete.

The results demonstrate a systematic widening of the performance gap, particularly among men and young athletes. This widening gap aligns with a general decline in participation, evident in the 11.9% decrease in German Athletics Association (DLV) membership and the massive 26.6% loss in the 27- to 60-year-old age group (Deutscher Leichtathletik-Verband, n.d.b). This age group is the largest pool of coaches, and their decline is particularly noticeable in technical disciplines, which rely on more intensive coaching.

Another reason for the wider performance spread could be the decline in performance among the extended elite due to reduced financial support. As Breuer et al. (2021) point out, the extended elite faced greater financial difficulties during the COVID-19 pandemic, leading many athletes to retire. This stems from their greater reliance on external support, which often dried up during the pandemic, while national team athletes relied on centralized funding. As a result, this extended elite has been lost. This lack of internal competition weakens international competitiveness in the long term, true to the saying ”competition is good for business”.

Our findings must to be understood in light of certain limitations. Because we focused exclusively on the top 35 athletes, our data represents the extended elite and may not apply to grassroots sports. Broader research should expand this scope, which would allow for a deeper analysis of underlying structural changes across all levels of performance. The trend toward polarization could explain the lack of medals at international championships. In our opinion, the focus should be on strengthening the extended elite and coach training to revitalize competition, which will also benefit elite athletes.

Contribution Statement

All authors jointly developed the research idea and collaborated closely throughout the project. Maximilian Wernz mainly contributed to data preparation, visualization, and repository management. Erik Müller contributed to data preparation and writing. Mattis Dietrich focused on the performance gap analyses, while Luca Bouché concentrated on the analysis of the COVID-19 pandemic.

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